Practical No. 3

AIM: Installation and Configuration of Virtualization using KVM.

Setting up Virtualization with KVM:

1. Check Hardware Support: Ensure that your CPU supports hardware virtualization (Intel VT-x or AMD-V). You may need to enable virtualization in your BIOS/UEFI settings.

2. Install KVM Packages:
On Ubuntu:
- sudo apt update
- sudo apt install qemu-kvm libvirt-clients libvirt-daemon-system bridge-utils

3. Start and enable the virtualization service:
- bash sudo systemctl enable --now libvirtd

4. Add your current user to the kvm and libvirt groups:
- bash sudo usermod -aG libvirt $USER sudo usermod -aG kvm $USER

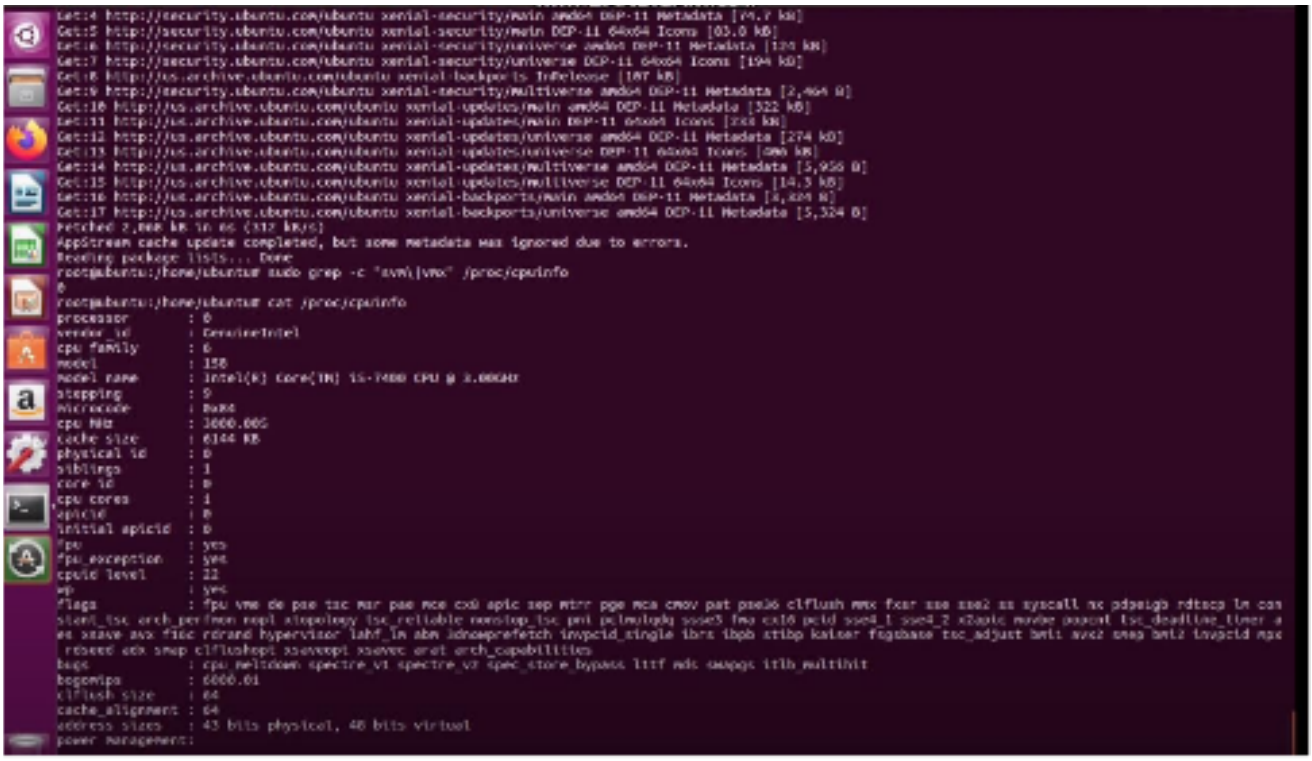
5. Log out and log back in (or restart your computer) so the new group permissions apply.
You can then open *Virtual Machine Manager* from your application menu to start creating virtual machines.

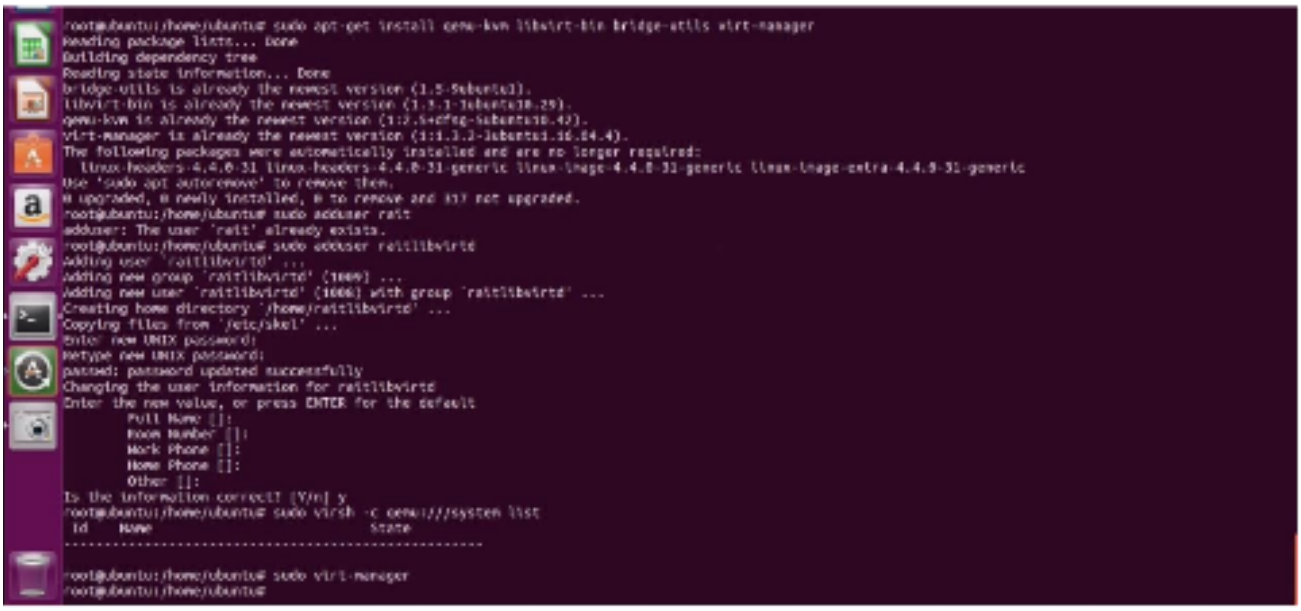
Steps to Create Virtual Machines:

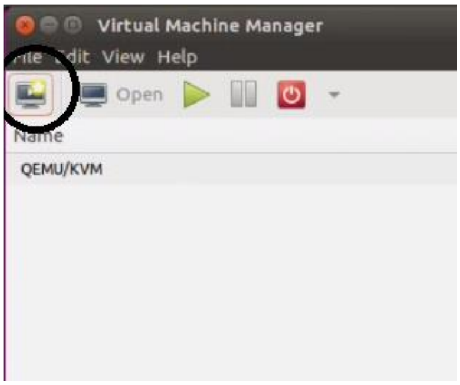
1. Create a Virtual Machine:
- Go to the "Virtual Machines" tab.
- Click "New VM" to create a new virtual machine.
- Follow the wizard to configure the VM, including specifying the operating system, memory, CPU, and storage settings.

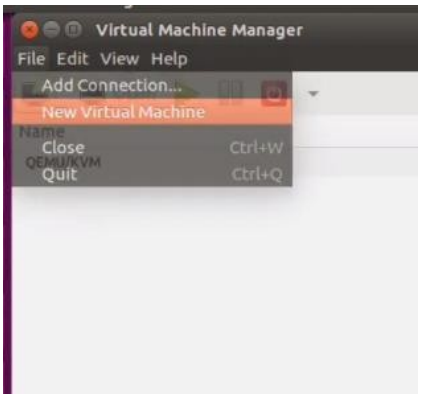
2. Install Operating System:
- Attach an ISO or a CD/DVD to the virtual machine for the OS installation.
- Power on the VM and access the console to install the operating system.

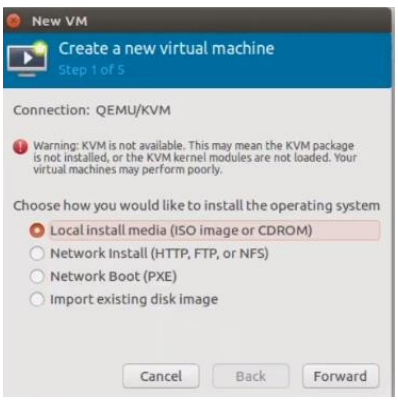
3. Manage Virtual Machines:
- From the "Virtual Machines" tab, you can start, stop, pause, or remove virtual
- machines.
- Monitor VM performance and access consoles for troubleshooting.

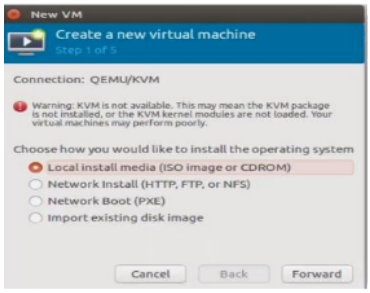

Get:4 http://security.ubuntu.com/ubuntu xenial-security/main amd64 DEP-11 Metadata [74.7 kB]
Get:5 http://security.ubuntu.com/ubuntu xenial-security/main DEP-11 64x64 Icons [83.0 kB]
Get:6 http://security.ubuntu.com/ubuntu xenial-security/universe amd64 DEP-11 Metadata [139 kB]
Get:7 http://security.ubuntu.com/ubuntu xenial-security/universe DEP-11 64x64 Icons [194 kB]
Get:8 http://us.archive.ubuntu.com/ubuntu xenial-backports InRelease [107 kB]
Get:9 http://security.ubuntu.com/ubuntu xenial-security/multiverse amd64 DEP-11 Metadata [2,464 B]
Get:10 http://us.archive.ubuntu.com/ubuntu xenial-updates/main amd64 DEP-11 Metadata [322 kB]
Get:11 http://us.archive.ubuntu.com/ubuntu xenial-updates/main DEP-11 64x64 Icons [288 kB]
Get:12 http://us.archive.ubuntu.com/ubuntu xenial-updates/universe amd64 DEP-11 Metadata [274 kB]
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Get:14 http://us.archive.ubuntu.com/ubuntu xenial-updates/multiverse amd64 DEP-11 Metadata [5,956 B]
Get:15 http://us.archive.ubuntu.com/ubuntu xenial-updates/multiverse DEP-11 64x64 Icons [14.5 kB]
Get:16 http://us.archive.ubuntu.com/ubuntu xenial-backports/main amd64 DEP-11 Metadata [4,204 B]
Get:17 http://us.archive.ubuntu.com/ubuntu xenial-backports/universe amd64 DEP-11 Metadata [5,324 B]
Fetched 2,008 kB in 6s (317 kB/s)
AppStream cache update completed, but some metadata was ignored due to errors.
Reading package lists... Done
root@ubuntu:/home/ubuntu# sudo grep -c "svm\|vmx" /proc/cpuinfo
8
root@ubuntu:/home/ubuntu# cat /proc/cpuinfo
processor : 0
vendor_id : GenuineIntel
cpu family : 6
model : 158
model name : Intel(R) Core(TM) i5-7400 CPU @ 3.00GHz
stepping : 9
microcode : 0x84
cpu MHz : 3000.006
cache size : 6144 KB
physical id : 0
siblings : 1
core id : 0
cpu cores : 1
apicid : 0
initial apicid : 0
fpu : yes
fpu_exception : yes
cpuid level : 22
wp : yes
flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon nopl xtopology tsc_reliable nonstop_tsc pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch invpcid_single ibrs ibpb stibp kaiser fsgsbase tsc_adjust bmi1 avx2 smep bmi2 invpcid mpx rdseed adx smap clflushopt xsaveopt xsavec xgetbv1 arch_capabilities
bugs : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs itlb_multihit
bogomips : 6000.01
clflush size : 64
cache_alignment : 64
address sizes : 43 bits physical, 48 bits virtual
power management:

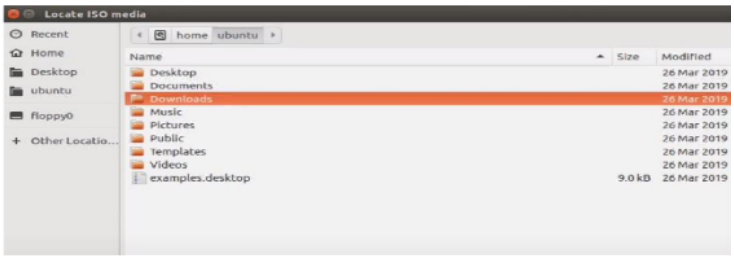

root@ubuntu:/home/ubuntu# sudo apt-get install qemu-kvm libvirt-bin bridge-utils virt-manager
Reading package lists... Done
Building dependency tree
Reading state information... Done
bridge-utils is already the newest version (1.5-9ubuntu1).
libvirt-bin is already the newest version (1.3.1-1ubuntu10.29).
qemu-kvm is already the newest version (1:2.5+dfsg-5ubuntu10.42).
virt-manager is already the newest version (1:1.3.2-3ubuntu1.16.04.4).
The following packages were automatically installed and are no longer required:
linux-headers-4.4.0-31 linux-headers-4.4.0-31-generic linux-image-4.4.0-31-generic linux-image-extra-4.4.0-31-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 117 not upgraded.
root@ubuntu:/home/ubuntu# sudo adduser ratt
adduser: The user 'ratt' already exists.
root@ubuntu:/home/ubuntu# sudo adduser rattlibvirtd
Adding user 'rattlibvirtd' ...
Adding new group 'rattlibvirtd' (1000) ...
Adding new user 'rattlibvirtd' (1006) with group 'rattlibvirtd' ...
Creating home directory '/home/rattlibvirtd' ...
Copying files from '/etc/skel' ...
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
Changing the user information for rattlibvirtd
Enter the new value, or press ENTER for the default
Full Name []:
Room Number []:
Work Phone []:
Home Phone []:
Other []:
Is the information correct? [Y/n] y
root@ubuntu:/home/ubuntu# sudo virsh -c qemu:///system list
Id Name State
----
root@ubuntu:/home/ubuntu# sudo virt-manager
root@ubuntu:/home/ubuntu#

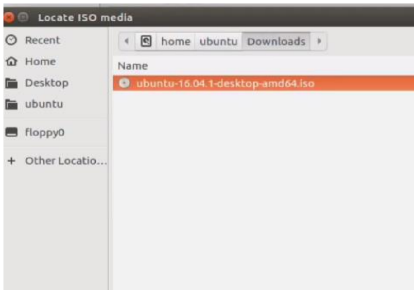

Virtual Machine Manager
File Edit View Help
Open
Name
QEMU/KVM

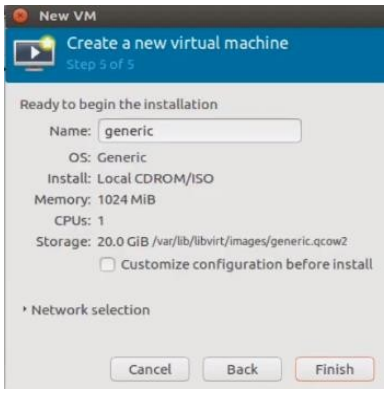

Virtual Machine Manager
File Edit View Help
Add Connection...
New Virtual Machine
Close Ctrl+W
Quit Ctrl+Q
Name
QEMU/KVM

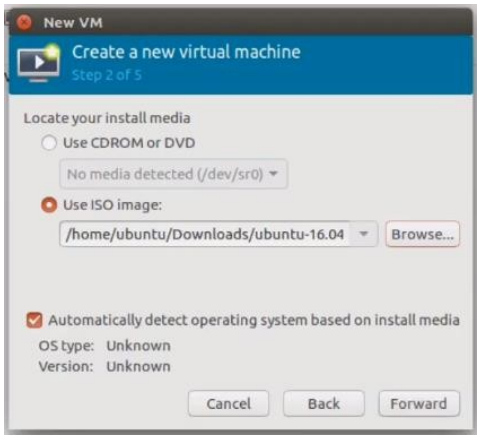

New VM
Create a new virtual machine
Step 1 of 5
Connection: QEMU/KVM
Warning: KVM is not available. This may mean the KVM package is not installed, or the KVM kernel modules are not loaded. Your virtual machines may perform poorly.
Choose how you would like to install the operating system
Local install media (ISO image or CDROM)
Network Install (HTTP, FTP, or NFS)
Network Boot (PXE)
Import existing disk image
Cancel
Back
Forward

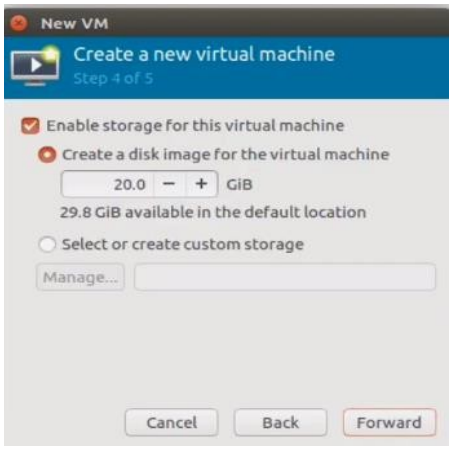

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Import existing disk image
Cancel
Back
Forward

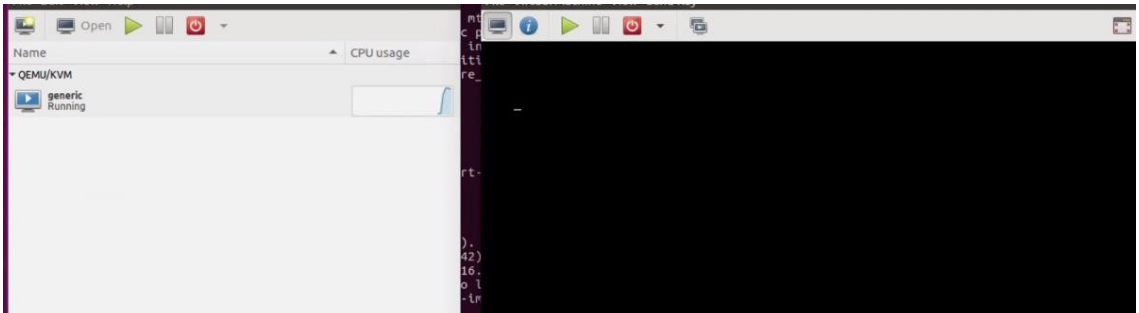

Locate ISO media
Recent
Home
Desktop
ubuntu
floppy0
Other Locatio...
home
ubuntu
Name
Size
Modified
Desktop
Documents
Downloads
Music
Pictures
Public
Templates
Videos
examples.desktop
9.0 kB
26 Mar 2019


Locate ISO media
Recent
Home
Desktop
ubuntu
floppy0
Other Locatio...
home
ubuntu
Downloads
Name
ubuntu-16.04.1-desktop-amd64.iso


New VM
Create a new virtual machine
Step 5 of 5
Ready to begin the installation
Name: generic
OS: Generic
Install: Local CDROM/ISO
Memory: 1024 MiB
CPUs: 1
Storage: 20.0 GiB /var/lib/libvirt/images/generic.qcow2
Customize configuration before install
Network selection
Cancel
Back
Finish


New VM
Create a new virtual machine
Step 2 of 5
Locate your install media
Use CDROM or DVD
No media detected (/dev/sr0)
Use ISO image:
/home/ubuntu/Downloads/ubuntu-16.04
Browse...
Automatically detect operating system based on install media
OS type: Unknown
Version: Unknown
Cancel
Back
Forward


New VM
Create a new virtual machine
Step 4 of 5
Enable storage for this virtual machine
Create a disk image for the virtual machine
20.0
GiB
29.8 GiB available in the default location
Select or create custom storage
Manage...
Cancel
Back
Forward


Open
Name
CPU usage
QEMU/KVM
generic
Running